STATISTICAL INFERENCE (STA 121)

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Students should be able to

- Define Statistics.
- Understand Different Branches of Statistics.
- Define and Understand Population and Sample.
- Understand Statistical Estimation Theory.
- Understand Statistical Hypothesis Testing.
- Understand Regression Analysis.
- Understand Correlation Theory.
- Understand Elementary Time Series.

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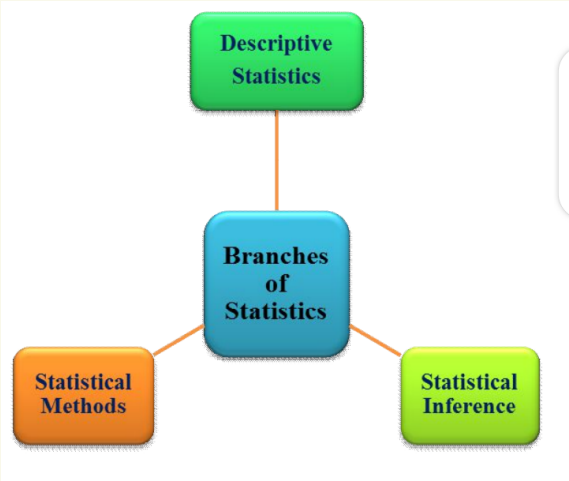
Objectives: Students should be able to

- Define Statistics
- Understand Different Branches of Statistics
- Define and Understand Population and Sample

Meaning of Statistics

- Statistics can simply be defined as the "science of data".
- It is the science of collection, organization and interpretation of numerical facts, which we called data.

Branches of Statistics



Branches of Statistics

Descriptive Statistics

Collection, summarization and presentation of numerical information in form of reports, charts and diagram.

Statistical Method

A device for classifying data and making clear relationship between variable under consideration, using the statistical tools and formulae.

Inferential Statistics

Makes use of information from a sample to draw conclusions (inferences) about the population from which the sample was taken.

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Population

Group of individuals or items to whom the conclusions of a study or experiment apply.

Finite Population

A population is said to be finite if it consists of a finite or fixed number of elements (items, objects, and measurements or observations).

Infinite Population

A population is said to be infinite if there is (at least hypothetically) no limit to the number of elements it can contain. For example, a possible roll of a pair of dice is an infinite population for there is no limit to the number of times they can be rolled.



Sample

A representative part of a population which can be random or purposive.

Advantages of sampling

- Low cost of sampling.
- Less time consuming in sampling.
- Scope of sampling is high.
- Accuracy of data is high.
- Organization of convenience.
- Intensive and exhaustive data.
- Suitable in limited resources.
- Better rapport.

Disadvantages of sampling

- Chances of bias
- Difficulties in selecting a truly representative sample
- Inadequate knowledge in the subject.
- Changeability of units.
- Impossibility of sampling.

Practice Question

1. What is the meaning of Statistics?
2. Discuss the types of Statistics?
3. Explain population.

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Objectives: Students should be able to

- Explain the Sampling Concept in Statistics.
- Understand the Probability and Non-Probability Sampling Techniques.
- Understand the Sampling Distribution of Means and Proportion and apply it to solve problems.

Sampling Techniques

Sampling is concerned with the selection of a subset of individuals from within a statistical population to estimate characteristics of the whole population.

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Reasons for Sampling

- It may be too expensive or time consuming to measure every item.
- It may be more accurate to measure a few items carefully, than to try to measure every item.
- It is essential to sample if by examining items you destroy them.

The disadvantage of sampling is that information is inevitably lost by not measuring every item.

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Probability and Non Probability Sampling

Probability/Random Sampling

Sampling technique in which sample from a larger population are chosen using a method based on the theory of probability. Examples are Simple Random Sampling, Systematic Sampling, Cluster Sampling, Stratified Sampling and Multistage Sampling

Non Probability Sampling

Sampling technique in which the researcher selects samples based on the subjective judgment of the researcher rather than random selection. Examples are Quota Sampling and Purposive Sampling.

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Sampling With Replacement

Sampling where each member of the population may be chosen more than once.

Sampling Without Replacement

Sampling where each member of the population cannot be chosen more than once.

Sampling Distribution

This is the computation of a statistic, which vary from sample to sample by considering all the possible samples of size N that can be drawn from a given population (either with or without replacement)

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Suppose that all possible samples of size n are drawn without replacement from a finite population of size N. If we denote the sampling mean and standard deviation of the sampling distribution of means by $\mu_{\bar{x}}$ and $\sigma_{\bar{x}}$ and the population mean and standard deviation by μ and σ, respectively, then

$$\mu_{\bar{x}} = \mu \text{ and } \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$$

If the population is infinite or if sampling is with replacement, the above results reduce to

$$\mu_{\bar{x}} = \mu \text{ and } \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

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Suppose that a population is infinite and that the probability of occurence of an event (called its success) is p, while the probability of nonoccurence of the event is $q = 1 - p$. The sampling distribution of proportions whose mean μ_p and standard deviation σ_p when sampling with replacement for a finte population are given by

$$\mu_p = p \text{ and } \sigma_p = \sqrt{\frac{pq}{n}} = \sqrt{\frac{p(1-p)}{n}}$$

For large values of n ($n \geq 30$), the sampling distribution is closely normally distributed. Note that the population is binomially distributed.

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Sampling Distribution of Differences and Sums

Given two populations, for sample sizes n_1 and n_2 drawn from different populations, we compute statistic S_1 and S_2 respectively which yields a sampling distribution for the statistics whose mean and standard deviation is denoted by μ_{S1}, μ_{S2} and standard deviation σ_{S1}, σ_{S2}. From all possible combinations of these samples from the two populations, we obtain a distribution of the differences $S_1 - S_2$, which is calledthe sampling distribution of difference of the statistics. The mean and standard deviation of this sampling distribution, denoted by μ_{S1-S2} and σ_{S1-S2}, are given by

$$\mu_{S1-S2} = \mu_{S1} - \mu_{S2} \text{ and } \sigma_{S1-S2} = \sqrt{\sigma_{S1}^2 + \sigma_{S2}^2}$$

provided the samples are independent

Sampling Distribution of Differences of Means

If S_1 and S_2 are the sample means from the two populations by denoting with $\bar{x}_1$ and $\bar{x}_2$ respectively, then the sampling distribution of the differences of means is

$$\mu_{\bar{x}_1-\bar{x}_2} = \mu_{\bar{x}_1} - \mu_{\bar{x}_2} = \mu_1 - \mu_2 \text{ and } \sigma_{\bar{x}_1-\bar{x}_2} = \sqrt{\sigma^2_{\bar{x}_1} + \sigma^2_{\bar{x}_2}} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Sampling Distribution of Differences of Proportion

Correspondingly, for two binomially distributed populations with parameters (p_1, q_1) and (p_2, q_2), the sampling distribution of the differences of proportion is given by

$$\mu_{p_1-p_2} = \mu_{p_1} - \mu_{p_2} = p_1 - p_2 \text{ and } \sigma_{p_1-p_2} = \sqrt{\sigma^2_{p_1} + \sigma^2_{p_2}} = \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$$

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If n_1 and n_2 are large ($n_1, n_2 \geq 30$), the sampling distributions of differences of means or proportions are closely normally distributed.

Sampling Distribution of Sums of Statistics

The mean and standard deviation of the distribution is given by

$$\mu_{S_1+S_2} = \mu_{S_1} - \mu_{S_2} \text{ and } \sigma_{S_1+S_2} = \sqrt{\sigma_{S_1}^2 + \sigma_{S_2}^2}$$

assuming the samples are independent.

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Illustration 1

A population consists of the five numbers 2, 3, 6, 8 and 11. Consider all possible samples of size 2 that can be drawn with replacement from this population. Find (a) the mean of the population, (b) the standard deviation of the population, (c) the mean of the sampling distribution of means and (d) the standard deviation of the sampling distribution of means (i.e., the standard error of means).

Solution

a) The Population Mean

$$\mu = \frac{\sum X_i}{N} = \frac{2+3+6+8+11}{5} = \frac{30}{5} = 6.0$$

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b) The Population Standard Deviation

$$\sigma^2 = \frac{\sum (X_i - \mu)^2}{N} = \frac{(2-6)^2 + (3-6)^2 + (6-6)^2 + (8-6)^2 + (11-6)^2}{5}$$

$$= \frac{16 + 9 + 0 + 4 + 25}{5} = 10.8$$

$$\sigma = 3.29$$

c) There are 5(5) = 25 samples of size 2 that can be drawn with replacement. These are

(2,2) (2,3) (2,6) (2,8) (2,11) (3,2) (3,3) (3,6) (3,8)
(3,11) (6,2) (6,3) (6,6) (6,8) (6,11) (8,2) (8,3) (8,6)
(8,8) (8,11) (11,2) (11,3) (11,6) (11,8) (11,11)

The corresponding sample means are

2.0, 2.5, 4.0, 5.0, 6.5, 2.5, 3.0, 4.5, 5.5, 7.0, 4.0, 4.5, 6.0, 7.0, 8.5, 5.0, 5.5, 7.0, 8.0, 9.5, 6.5, 7.0, 8.5, 9.5, 11.0.

The mean of sampling distribution of mean is

$$\mu_{\bar{X}} = \frac{\text{sum of all sample means above}}{25} = \frac{150}{25} = 6.0$$

illustrating the fact that $\mu_{\bar{X}} = \mu$

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d) The variance $\sigma_{\bar{x}}^2$ of the sampling distribution is obtained by deducting the mean 6 from each of the sample means, squaring the result and adding together and divide the obtained value by 25.

$$\sigma_{\bar{x}}^2 = \frac{135}{25}$$

$$\text{Thus} \sigma_{\bar{x}} = \sqrt{5.40} = 2.32$$

This illustrates the fact that for finite populations involving sampling with replacement (or infinite populations), $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$

Illustration 2

Solve Illustration 1 for the case that the sampling is without replacement.

Solution

As in parts (a) and (b) in Illustration 1, $\mu = 6$ and $\sigma = 3.29$.

(c) There are $\binom{10}{2} = 10$ samples of size 2 that can be drawn without replacement from the population: (2,3), (2,6), (2,8), (2,11), (3,6), (3,8), (3,11), (6,8), (6,11) and (8,11). The corresponding sample means are 2.5, 4.0, 5.0, 6.5, 4.5, 5.5, 7.0, 7.0, 8.5 and 9.5 and the mean of sampling distribution of means is

$$\mu_{\bar{X}} = \frac{2.5 + 4.0 + 5.0 + 6.5 + 4.5 + 5.5 + 7.0 + 7.0 + 8.5 + 9.5}{10} = 6.0$$

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(d) The variance of sampling distribution of means is

$$\sigma_{\bar{x}}^2 = \frac{(2.5-6.0)^2 + (4.0-6.0)^2 + (5.0-6.0)^2 + ... + (9.5-6.0)^2}{10} = 4.05$$

and $\sigma_{\bar{x}} = 2.01$. This illustrates

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right)$$

since the right side equals

$$\frac{10.8}{2}\left(\frac{5-2}{5-1}\right) = 4.05$$

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Illustration 3

Assume that the heights of 3000 soccer players in a tournament are normally distributed with mean 68.0 inches (in) and standard deviation 3.0in. If 80 samples consisting of 25 players each are obtained, what would be the expected mean and standard deviation of the resulting sampling distribution of means if the sampling were done (a) with replacement and (b) without replacement?

Solution

(a)

$$\mu_{\bar{x}} = \mu = 68.0in \quad \text{and} \quad = \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{25}} = 0.6in$$

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(b)

$$\mu_{\bar{x}} = 68.0in \quad \text{and} \quad = \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}} = \frac{3}{\sqrt{25}}\sqrt{\frac{3000-25}{3000-1}} = 0.598in$$

which is slightly less than 0.6 in.

Illustration 4

In how many samples of Illustration 3 would you expect to find the mean between (a) between 66.8 and 68.3 in and (b) less than 66.4 in?

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Solution

The mean $\bar{x}$ of a sample in standard units is here given by

$$z = \frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{\bar{x} - 68.0}{0.6}$$

(a)

$$P(66.8 \leq \bar{X} \leq 68.3)$$

$$P\left(\frac{66.8 - 68.0}{0.6} \leq Z \leq \frac{68.3 - 68.0}{0.6}\right)$$

$$= \phi\left(\frac{68.3 - 68.0}{0.6}\right) - \phi\left(\frac{66.8 - 68.0}{0.6}\right)$$

$$= \phi(0.5) - \phi(-2.0)$$

$$= \phi(0.5) - 1 + \phi(2.0)$$

$$= 0.6915 + 0.9772 - 1 = 0.6687$$

Therefore, the expected number of samples is $(80)(0.6687) = 53$

(b)

$$P(\bar{x} \leq 66.4)$$

$$P\left(Z \leq \frac{66.4 - 68.0}{0.6}\right)$$

$$= \phi\left(\frac{66.4 - 68.0}{0.6}\right)$$

$$= \phi(-2.67) = 1 - \phi(2.67)$$

$$= 1 - 0.9962 = 0.0038$$

Thus, the expected number of samples is $(80)(0.0038) = 0$

Illustration 5

Find the probability that in 120 tosses of a fair coin (a) less than 40% or more than 60% will be heads and (b) $\frac{5}{8}$ or more will be heads

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Solution

We consider the 120 tosses of the coin to be a sample from the infinite population of all possible tosses of the coin. In this population the probability of heads is $p = \frac{1}{2}$ and the probability of tails is $q = 1 - p = \frac{1}{2}$

(a) Using normal approximation to binomial we require that the number of heads in 120 tosses will less than 48 or more than 72. Since the number of heads is a discrete variable, we ask for the probability that the number of heads is less than 47.5 or greater than 72.5.

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$$\mu = \text{expected number of heads} = Np = 60 \quad \text{and}$$

$$\sigma = \sqrt{Npq} = \sqrt{(120)(\frac{1}{2})(\frac{1}{2})} = 5.48$$

$$P\left(Z \leq \frac{47.5 - 60}{5.48}\right) \text{or} \quad P\left(Z \geq \frac{72.5 - 60}{5.48}\right)$$

$$= \phi\left(\frac{47.5 - 60}{5.48}\right) + 1 - \phi\left(\frac{72.5 - 60}{5.48}\right)$$

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$$= \phi(-2.28) + 1 - \phi(2.28)$$
$$= 1 - \phi(2.28) + 1 - \phi(2.28)$$
$$= 2 - (\phi(2.28) + \phi(2.28))$$
$$= 2 - 2(0.9887) = 2 - 1.9774 = 0.0226$$

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Illustration 6

The solar light bulbs of company K have a mean life time of 1400 hours (h) with a standard deviation of 200 h, while those of company L have a mean lifetime of 1200 h with a standard deviation of 100 h. If random samples of 125 bulbs of each brand are tested, what is the probability that the brand K bulbs will have a mean life time that is at least (a) 160 h and (b) 250 h more than the brand L bulbs?

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Solution

Let $\bar{x}_K$ and $\bar{x}_L$ denote the mean lifetimes of samples K and L, respectively, Then

$$\mu_{\bar{x}_K - \bar{x}_L} = \mu_{\bar{x}_K} - \mu_{\bar{x}_L} = 1400 - 1200 = 200h$$

and

$$\sigma_{\bar{x}_K - \bar{x}_L} = \sqrt{\frac{\sigma_K^2}{n_K} + \frac{\sigma_L^2}{n_L}} = \sqrt{\frac{(100)^2}{125} + \frac{(200)^2}{125}} = 20h$$

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The standardized variable for the difference in mean is

$$z = \frac{(\bar{x}_K - \bar{x}_L) - (\mu_{\bar{x}_K - \bar{x}_L})}{\sigma_{\bar{x}_K - \bar{x}_L}} = \frac{(\bar{x}_K - \bar{x}_L) - 200}{20}$$

and is very closely normally distributed.
(a) The difference in 160 h is $P[(\bar{x}_K - \bar{x}_L) \geq 160] = 1 - P[(\bar{x}_K - \bar{x}_L) \leq 160]$

$$1 - P\left(z \leq \frac{160 - 200}{20}\right)$$

$$1 - \phi\left(\frac{160 - 200}{20}\right)$$

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$$1 - \phi(-2) = 1 - 1 + \phi(2) = 0.9772$$

(b) The difference in 250 h is $P[(\bar{x}_K - \bar{x}_L) \geq 250]$.

$$1 - P\left(z \leq \frac{250 - 200}{20}\right)$$

$$1 - \phi\left(\frac{250 - 200}{20}\right)$$

$$1 - \phi(2.50) = 1 - 0.9938 = 0.0062$$

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Objectives: Students should be able to

- Understand the Concept of Point and Interval Estimates.
- Solve problems on Estimation Theory.

Estimation

Estimation is basically a process by which the sample statistics obtained is used to estimate the parameters of the population from which the sample was drawn.

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Point and Interval Estimates

An estimate of a population parameter given by a single number is called a point estimate of the parameter.An estimate of a population parameter given by two numbers between which the parameter may be considered to lie is called an interval estimate of the parameter or the confidence interval. A confidence interval gives an idea of how close the true value of the parameter might be to the point estimate. A statement of the error (or precision) of an estimate is often called its reliability.

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Unbiased Estimates

If the mean of the sampling distribution of a statistic equals the corresponding population parameter, the statistic is called an unbiased estimator of the parameter; otherwise, it is called a biased estimator.

Efficient Estimates

If the sampling distributions of two statistics have the same mean (or expectation), then the statistic with the smaller variance is called an efficient estimator of the mean, while the other statistic is called an inefficient estimator.

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Confidence Interval for Means

If the statistics S is the sample mean $\bar{x}$, then the 95% and 99% confidence limits for estimating the population mean μ are given by $\bar{x} \pm 1.96\sigma_{\bar{x}}$ and $\bar{X} \pm 2.58\sigma_{\bar{x}}$, respectively. More generally, the confidence limits are given by $\bar{x} \pm z_{\frac{\alpha}{2}}\sigma_{\bar{x}}$

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Sampling from infinite population or with replacement from a finite population

The confidence limit is given by

$$\bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

Sampling without replacement fom a finite population of size N

The confidence limit is given by

$$\bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

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Sampling from infinite population or with replacement from a finite population

The confidence limit is given by

$$P \pm z_{\frac{\alpha}{2}} \sqrt{\frac{pq}{n}} = P \pm z_{\frac{\alpha}{2}} \sqrt{\frac{p(1-p)}{n}}$$

Sampling without replacement fom a finite population of size N

The confidence limit is given by

$$P \pm z_{\frac{\alpha}{2}} \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}}$$

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Difference of Two Population Means when Populations are infinite

The confidence limit is given by

$$\bar{x}_1 - \bar{x}_2 \pm z_{\frac{\alpha}{2}} \sigma_{\bar{x}_1 - \bar{x}_2} = \bar{x}_1 - \bar{x}_2 \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Difference of Two Population Proportions when Populations are infinite

The confidence limit is given by

$$p_1 - p_2 \pm z_{\frac{\alpha}{2}} \sigma_{p_1 - p_2} = p_1 - p_2 \pm z_{\frac{\alpha}{2}} \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

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Illustration 1

In a sample of five measurements, the diameter of a sphere was recorded by a student in a laboratory as 6.33, 6.37, 6.36, 6.32, and 6.37 centimeters (cm). Determine unbiased and efficient estimates of (a) the true mean and (b) the true variance.

Solution

(a) The unbiased and efficient estimate of the true mean (i.e., the populations mean) is

$$\hat{X} = \frac{\sum X}{N} = \frac{6.33 + 6.37 + 6.36 + 6.32 + 6.37}{5} = 6.35cm$$

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(b) The unbiased and efficient estimate of the true variance (i.e., the population variance) is

$$\hat{s}^2 = \frac{N}{N-1}s^2 = \frac{\sum(X-\bar{X})^2}{N-1}$$

$$= \frac{(6.33-6.35)^2 + (6.37-6.35)^2 + ... + (6.37-6.35)^2}{5-1}$$

$$= 0.00055cm^2$$

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Illustration 2

The standard deviation of bulbs manufactured by AYZ Solar Company is 5.6. If the mean life span of 64 bulbs were randomly selected from the lot is 60 days.

(i) construct the 95% confidence limit for the bulb

(ii) what is the minimum number of samples to be selected so that the error does not exceed 0.5

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Solution

(i) $\sigma = 5.6$, $\bar{x} = 60$, $n = 64$, $\alpha = 95\% = 0.95$, $1 - 0.95 = 0.05$, Then, $\frac{0.05}{2} = 0.025$
From normal distribution table $z_{\frac{\alpha}{2}} = 1.96$.
The confidence interval, $C.I. = \bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$

$$= 60 \pm 1.96 \times \frac{5.6}{\sqrt{64}}$$

$$= 60 \pm 1.372$$

$$= [60 - 1.372, 60 + 1.372]$$

$$= [58.628, 61.372]$$

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(ii) For Error ≤ 0.5

$$z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq 0.5$$

$$\frac{1.96 \times 5.6}{\sqrt{n}} \leq 0.5$$

$$1.96 \times 5.6 \leq 0.5\sqrt{n}$$

$$\frac{1.96 \times 5.6}{0.5} \leq \sqrt{n}$$

$$21.952 \leq \sqrt{n}$$

$$(\sqrt{n})^2 \geq (21.952)^2$$

$$n \geq 482$$

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Therefore, for the error not to exceed 0.5, the minimum number of samples to be selected is 482.

Illustration 3

An animal scientist studying the effect of new substance added to the diet of chinchila rabbits on the weights over a month period. The result of the effect on 7 rabbits choosing as a sample are shown in the table below.

Original Weights (W_o)	56.1	22.2	50.1	39.5	10.3	20.2	7.4
New Weights(W_n)	99.1	52.3	87.4	78.2	68.2	86.9	29.5

Find a 95% symmetric confidence interval for the weight gained. If the distribution is assumed to be normally distributed.

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Solution

W_o	W_n	$x = W_n - W_o$	x^2
56.1	99.5	43.4	1883.56
22.2	52.3	30.1	906.01
50.1	87.4	37.3	1391.29
39.5	78.2	38.7	1497.69
30.3	68.2	37.9	1436.41
40.2	86.9	46.7	2180.89
7.4	29.5	22.1	488.41
Total		256.2	9784.26

where, x is the gain in weight over a period of 1 month. The distribution is normal, the variance is unknown and $n = 7(< 30)$, then we use t value instead of Z value.

Sample mean

$$\bar{x} = \frac{\sum x}{n} = \frac{256.2}{7} = 36.6$$

Sample variance

$$s^2 = \frac{\sum X^2 - \frac{(\sum X)^2}{n}}{n-1} = \frac{9784.26 - \frac{(256.2)^2}{7}}{7-1} = 67.89$$

$$s = 8.24$$

t value from the table

$$t_{\alpha,n-1} = t_{0.025,6} = 2.447$$

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Confidence Interval

$$= \bar{x} \pm t_{\alpha} \frac{s}{\sqrt{n}}$$

$$= 36.6 \pm 2.447 \times \frac{8.24}{\sqrt{7}}$$

$$= 36.6 \pm 7.62$$

$$= [36.6 - 7.62, 36.6 + 7.62]$$

$$= [28.98, 44.22]$$

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Illustration 4

A random sample of 50 Statistics grades out of a total of 200 showed a mean of 75 and a standard deviation of 10.

(i) What are the 95% confidence limits for estimates of the mean of the 200 grades?

(ii) With what degree of confidence could we say that the mean of all 200 grades is 75 ± 1?

Solution

(a) Since the population size is not very large compared with the sample size, we must adjust for it. Then the 95% confidence limits are

$$\bar{x} \pm 1.96\sigma_{\bar{X}}$$

$$= \bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

$$= 75 \pm 1.96 \frac{10}{\sqrt{50}} \sqrt{\frac{200-50}{200-1}}$$

$$= 75 \pm 2.4$$

$$= [75 - 2.4, 75 + 2.4]$$

$$= [72.6, 77.4]$$

(b) The confidence limit can be represented by

$$\bar{x} \pm z_c \sigma_{\bar{x}}$$

$$= \bar{x} \pm z_c \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

$$= 75 \pm z_c \frac{10}{\sqrt{50}} \sqrt{\frac{200-50}{200-1}}$$

$$= 75 \pm 1.23 z_c$$

Since this must equal to 75 ± 1, we have $1.23Z_c = 1$, or $z_c = 0.81$. The area under the normal curve from $z = 0$ to $z = 0.81$ is $0.7910 - 0.50 = 0.2910$, hence the required degree of confidence is $2(0.2910) = 0.582$, or 58.2%

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Hypothesis

A hypothesis is an idea that is based on known facts and is used for further reasoning or investigation.

Statistical Hypothesis

A statistical hypothesis is an assertion or conjecture concerning one or more populations which may be true or false.

Types of Hypothesis

Null Hypothesis (H_0): a statistical hypothesis that states no difference.
Alternate Hypothesis (H_1): a statistical hypothesis that states the existence of difference.

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Full Specification of Hypothesis Test

- H_0: $\mu = k$; H_1: $\mu < k$ (one sided) $\Longrightarrow$ lower tail or left tailed test
- H_0: $\mu = k$; H_1: $\mu > k$ (one sided) $\Longrightarrow$ upper tail or right tailed test
- H_0: $\mu = k$; H_1: $\mu \neq k$ (two sided) $\Longrightarrow$ two tail test

Very Important

Note that failure to reject H_0 does not mean the null hypothesis is true. There is no formal outcome that says "accept H_0." It only means that we do not have sufficient evidence to support H_1.

Statistical Test

This uses the data obtained from a sample to make a decision about whether the null hypothesis should be rejected.

Test Statistic

The numerical value obtained from a statistical test. Below are the distributions and their respective conditions

Case 1

Test for mean, known variance, normal distribution or large sample i.e. $n > 30$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

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Case 2

Test for mean, large sample, unknown variance.

$$Z = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

Case 3

Test for mean, unknown variance, small sample i.e. $n < 30$

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

with n-1 degree of freedom

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Case 4

Test for proportion, large sample

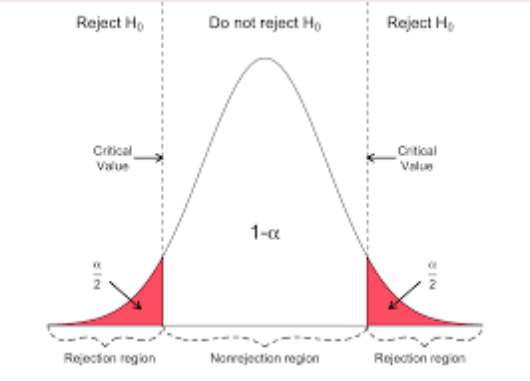
$$Z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

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Critical Region

This is a set on the real number line that leads to the rejection of H_0 in favour of H_1



Type I Error

This is the possibility of rejecting the null hypothesis when it is true

Type II Error

This is the possibility of not rejecting the null hypothesis when it is false

Result of a Statistical Test

	H_0 is true	H_1 is false
Reject H_0	Type I Error	Correct Decision
Do not reject H_0	Correct Decision	Type II Error

Procedure for Hypothesis Test

1. State the hypotheses (the null and the alternate).
2. Identify the claim by determining the appropriate test statistic to be used.
3. Determine the significance level of the test.
4. Find the critical value(s) from the appropriate table.
5. Decide on the distribution of the test statistics and the sidedness of the test whether one tailed or two tailed. Then compute the test value.
6. Decide the boundaries of the critical region, state your decision rule and make conclusion.

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Level of Significance

This is the maximum probability of committing Type I error. It is usually denoted by α. The probabiity of committing Type II error is denoted by β

Power of a test

This is the probability of rejecting H_0 given that the specific alternative hypothesis is true. That is Power = $1 - \beta$.

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Properties of Hypothesis Testing

- α and β are related; decreasing one generally increases the other.
- α can be set to a desired value by adjusting the critical value. Typically it is always set at 0.05 or 0.01.
- Increasing n decreases both α and β
- β decreases as the distance between the true value and hypothesized value (H_1) increases.

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Illustration 1

A random sample of size 35 selected from a population whose distribution is normal with mean μ and variance 36 gives a sample mean 48. Test the hypothesis, H_0: $\mu = 50$ against the sample mean 48, H_1: $\mu < 50$ at 5% level of significance.

Solution

$$H_0 : \mu = 50$$

$$H_1 : \mu < 50$$

Here, $n > 30$ i.e. n=35 and the variance is known. $\sigma^2 = 36 \Longrightarrow \sigma = 6$

$$Z_{calc} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{48 - 50}{6/\sqrt{35}}$$

$$= -1.972$$

$\alpha = 5\% = 0.05$, Since it is a lower tail test, $Z_{tab} = -1.645$

Conclusion: Since the test statistics Z_{calc} falls within the rejection region, the null hypothesis is rejected and thereby conclude that $\mu < 50$

Illustration 2

A quality control engineer finds that a sample of 100 light bulbs had an average life-time of 470 hours. Assuming a population standard deviation of $\sigma = 25 hours$, test whether the population mean is 480 hours against the alternative hypothesis $\mu < 480$ at a significance level of $\alpha = 0.05$

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Solution

$$H_0 : \mu = 480$$

$$H_1 : \mu < 480$$

$n = 100$ and the variance is known. $\sigma = 25$

$$Z_{calc} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{470 - 480}{25/\sqrt{100}}$$

$$= -4.0$$

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$\alpha = 5\% = 0.05$, Since it is a lower tail test, $Z_{tab} = -1.645$
Conclusion: Since the test statistics Z_{calc} falls within the rejection region, the null hypothesis is rejected and thereby conclude that $\mu < 480$

Illustration 3

The time taken to shave the hair from the head of people were recorded by a hair stylist. The mean time was found to be μ minutes and the standard deviation was σ For these three individuals, the time taken for shaving were 3.52, 5.40, 4.33, 3.20 and 2.50 minutes. Test at 5% significance whether the mean time is equal to 3.45 or not

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Solution

$$H_0 : \mu = 3.45$$

$$H_1 : \mu \neq 3.45$$

$n = 5$ (small sample), variance unknown.

$$\sum X = 3.52 + 5.40 + 4.33 + 3.20 + 2.50 = 18.95$$

$$\sum X^2 = 3.52^2 + 5.40^2 + 4.33^2 + 3.20^2 + 2.50^2 = 76.7893$$

$$\bar{X} = \frac{\sum X}{n} = \frac{18.95}{5} = 3.79$$

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$$s^2 = \frac{\sum X^2 - \frac{(\sum X)^2}{n}}{n-1}$$

$$= \frac{76.7893 - \frac{(18.95)^2}{5}}{5-1} = 1.2422$$

$$s = 1.1145$$

$$t_{calc} = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

$$= \frac{3.79 - 3.45}{1.1145/\sqrt{5}} = 0.6822$$

$\alpha = 5\% = 0.05$, Since it is a two tail test, $t_{tab} = t_{\frac{\alpha}{2},n-1} = t_{0.975,4} = 2.776$

Conclusion: Since the test statistics t_{calc} does not fall within the rejection region, we do not reject the null hypothesis and thereby conclude that $\mu = 3.45$

Illustration 4

A batch of 100 resistors have an average of 101.5 Ω. Assuming a population standard deviation of 5 Ω:

(a) Test whether the population mean is 100 Ω at 0.05 level of significance.

(b) Compute the p-value.

Solution

(a)

$$H_0 : \mu = 100$$

$$H_1 : \mu \neq 100$$

$n = 100$ and the variance is known. $\sigma = 5$

$$= \frac{101.5 - 100}{5/\sqrt{100}}$$

$$= 3.0$$

$\alpha = 5\% = 0.05$, Since it is a two tail test, $Z_{tab} = 1.96$
Conclusion: Since the test statistics Z_{calc} falls within the rejection region, the null hypothesis is rejected and thereby conclude that $\mu \neq 100$.
(b) Since the observed Z value is 3. Then, the p-value is

$$p = 2Pr(Z > 3) = 2 \times 0.00135 = 0.0027$$

This means that H_0 could have been rejected at sig nificance level $\alpha = 0.0027$ which is much stronger than rejecting it a 0.05.

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Illustration 5

An educator estimates that the dropout rate for seniors at high schools in Benin City is 12%. Last year in a random sample of 300 Benin City seniors, 27 withdrew from school. At $\alpha = 0.05$, is there enough evidence to reject the educators claim?

Solution

$$H_0 : p = 0.12$$

$$H_1 : p \neq 0.12$$

$n = 300$, $\hat{p} = \frac{27}{300} = 0.09$

$$Z_{calc} = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} = \frac{0.09 - 0.12}{\sqrt{\frac{0.12(1-0.12)}{300}}} = \frac{-0.03}{0.01876} = -1.60$$

$\alpha = 5\% = 0.05$, Since it is a two tail test, $Z_{tab} = 1.96$

Conclusion: Since the test statistics Z_{calc} falls within the non-critical region, we do not reject the null hypothesis and thereby do not have sufficient evidence to reject the claim that the rate for seniors at high schools in Benin City is 12%

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Definition

Regression analysis is defined as the analysis of relationships among variables. It is a statistical tool for the investigation of relationships between variables.

The general form of the equation is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + - - - + \beta_k X_k$$

where
Y = dependent or response variable
$X_1, X_2, ..., X_k$ are called the explanatory or independent variables
$\beta_0, \beta_1, ..., \beta_k$ are called the regression coefficients.

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Method of FInding the Regression Line

- Graphical Method (Scatter Diagram)
- Method of Least Square Line

The simple regression line is defined as

$$Y = \beta_0 + \beta_1 X$$

where

$$\beta_1 = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}$$

and

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

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Illustration 1

The table below shows the heights to the nearest inch (in) and the weights to the nearest pound (lb) of a sample of planks in a workshop.

X (in)	1	3	4	6	8	9	11	14
Y (lb)	1	2	4	4	5	7	8	9

(i) Construct a regression line for the data.

(ii) Find the weight of a plank whose height is 5 in.

(iii) Find the height of a plank whose weight is 6 lb.

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Solution

X	Y	X^2	XY
1	1	1	1
3	2	9	6
6	4	36	24
8	5	64	40
9	7	81	63
11	8	121	88
14	9	196	126
$\sum X = 52$	$\sum Y = 36$	$\sum X^2 = 508$	$\sum XY = 348$

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$$\bar{X} = \frac{\sum X}{n} = \frac{52}{7} = 7.43$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{36}{7} = 5.14$$

$$\beta_1 = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}$$

$$\beta_1 = \frac{7(348) - 52(36)}{7(508) - 52^2} = 0.662$$

$$\beta_0 = \bar{Y} - \beta_1 \bar{X} = 7.43 - 0.636(5.14) = 4.161$$

Then, the regression line is

$$Y = 4.161 + 0.662X$$

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(i) The weight of the plank whose height is 5 in

$$Y = 4.161 + 0.662(5)$$

$$Y = 4.161 + 3.310$$

$$= 7.471lb$$

when the height of the plank is 5 in, the weight of the plank is 7.471 lb.

(ii) The height of the plank whose weight is 6 lb

$$6 = 4.161 + 0.662(X)$$

$$X = \frac{1.839}{0.662} = 2.778in$$

when the weight of the plank is 6 lb, the height of the plank is 2.778 in.

Illustration 2

(a) Show that the equation of a straight line that passes through the points (X_1, Y_1) and (X_2, Y_2) is given by

$$Y - Y_1 = \frac{Y_2 - Y_1}{X_2 - X_1}(X - X_1)$$

(b) Find the equation of a straight line that passes through the points (2, -3) and (4, 5).

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Solution

(a) The equation of a straight line is

$$Y = \beta_0 + \beta_1 X \tag{1}$$

Since (X_1, Y_1) lies on the line,

$$Y_1 = \beta_0 + \beta_1 X_1 \tag{2}$$

Since (X_2, Y_2) lies on the line,

$$Y_2 = \beta_0 + \beta_1 X_2 \tag{3}$$

Subtracting equation (2) from (1),

$$Y - Y_1 = \beta_1 (X - X_1) \tag{4}$$

Subtracting equation (2) from (3),

$$Y_2 - Y_1 = \beta_1(X_2 - X_1) \quad \text{or} \quad \beta_1 = \frac{Y_2 - Y_1}{X_2 - X_1}$$

Substituting this value of β_1 into equation (4), we obtain

$$Y - Y_1 = \frac{Y_2 - Y_1}{X_2 - X_1}(X - X_1)$$

as required.
(b) Corresponding o the first point (2, -3), we have $X_1 = 2$ and $Y_1 = -3$; corresponding to the second point (4,5), we have $X_2 = 4$ and $Y_2 = 5$. Thus the slope is

$$\beta_1 = \frac{Y_2 - Y_1}{X_2 - X_1} = \frac{5 - (-3)}{4 - 2} = \frac{8}{2} = 4$$

and the required equation is

$$Y - Y_1 = \beta_1(X - X_1)$$

$$Y - (-3) = 4(X - 2)$$

which can be written as

$$Y = 4X - 11$$

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Practice Question

The table below gives experimental values of the pressure P of a given mass of gas corresponding to various values of the volume V. According to thermodynamic principles, a relationship having the form $PV^{\gamma} = C$, where γ and C are constants, should exists between the variables.

V (in^3)	54.3	61.8	72.4	88.7	118.6	194.0
P (lb/in^2)	61.2	49.2	37.6	28.4	19.2	10.1

(i) Find the values of γ and C

(ii) Write the equation connecting P and V

(iii) Estimate P when V = 100.0in^3

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Definition

Correlation refers to the mutual or degree of relationship between two or more variables.

Corrleation can be

- Perfect (Negative of Positive)
- Partial
- Zero (i.e. no correlation)

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Measurement of the Degree of Relationship

- Product Moment Correlation Coefficient.
- Spearman Rank Correlation Coefficient.

Pearson Product Moment Correlation Coefficient

$$r = \frac{n\sum XY - \sum X \sum Y}{\sqrt{[n\sum X^2 - (\sum X)^2][n\sum Y^2 - (\sum Y)^2]}} = \frac{S_{XY}}{\sqrt{S_{XX}S_{XY}}}$$

Spearman Rank Correlation Coefficient

$$r = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

d is the difference between the ranks
n is the number of obsevations.

Illustration 1

A study recorded the starting salary (in thousands), Y, and years of education, X, for 10 workers. The data is shown in the table below

Starting Salary	35	46	48	50	40	65	28	37	49	55
Years of Education	12	16	16	15	13	19	10	12	17	14

(i) Find the Product Moment Correlation Coefficient.

(ii) Spearman Rank Correlation Coefficient

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Solution (i)

Y	X	X^2	Y^2	XY
35	12	144	1225	420
46	16	256	2116	736
48	16	256	2304	768
50	15	225	2500	750
40	13	169	1600	520
65	19	361	4225	1235
28	10	100	784	280
37	12	144	1225	444
49	17	289	2401	833
55	14	196	3025	770
$\sum Y = 453$	$\sum X = 144$	$\sum X^2 = 2140$	$\sum Y^2 = 21549$	$\sum XY = 6756$

$$r = \frac{n\sum XY - \sum X \sum Y}{\sqrt{[n\sum X^2 - (\sum X)^2][n\sum Y^2 - (\sum Y)^2]}}$$

$$r = \frac{10(6756) - 144(453)}{\sqrt{[10(2140) - 144^2][10(21549) - 453^2]}}$$

$$= \frac{2328}{2612.773} = 0.89$$

Conclusion: It shows there is a strong positive correlation between the starting salary and years of education of the workers.

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Solution (ii)

Y	X	R_y	R_x	$d = R_y - R_x$	d^2
35	12	9	8.5	0.5	0.25
46	16	6	3.5	2.5	6.25
48	16	5	3.5	1.5	2.25
50	15	3	5	-2	4
40	13	7	7	0	0
65	19	1	1	0	0
28	10	10	10	0	0
37	12	8	8.5	-0.5	0.25
49	17	4	2	2	4
55	14	2	6	-4	16
					$\sum d^2 = 33$

$$r = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

$$r = 1 - \frac{6(33)}{10(10^2-1)}$$

$$= 1 - \frac{198}{10(99)}$$

$$= 1 - 0.20 = 0.80$$

Conclusion: It shows there is a strong positive correlation between the starting salary and years of education of the workers.

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Definitions

An ordered sequence of values of a variable at equally spaced time intervals.

Applications

- Economic Forecasting
- Sales Forecasting
- Budgetary Analysis
- Stock Market Analysis
- Yield Projections
- Process and Quality Control
- Inventory Studies
- Utility Studies
- Census Analysis etc

Components of Time Series

- Trend (T)
- Cyclical Variation (C)
- Seasonal Variation (S)
- Irregular Variation (I)

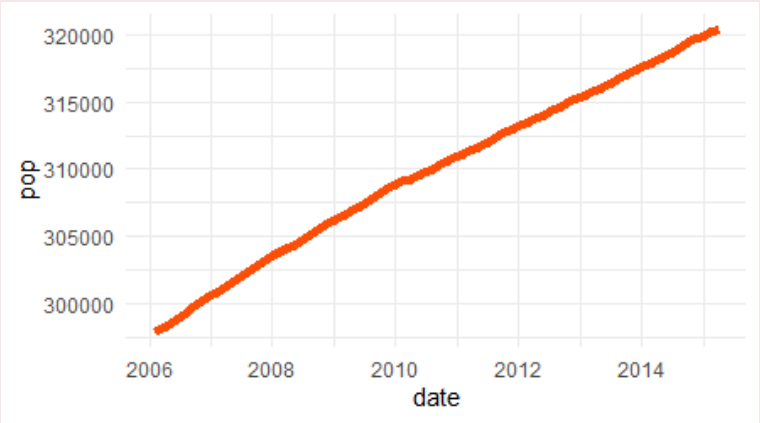
Trend

It refers to stationary, upward or downward movement that characterise a time series over a period of time.

Examples of Trend

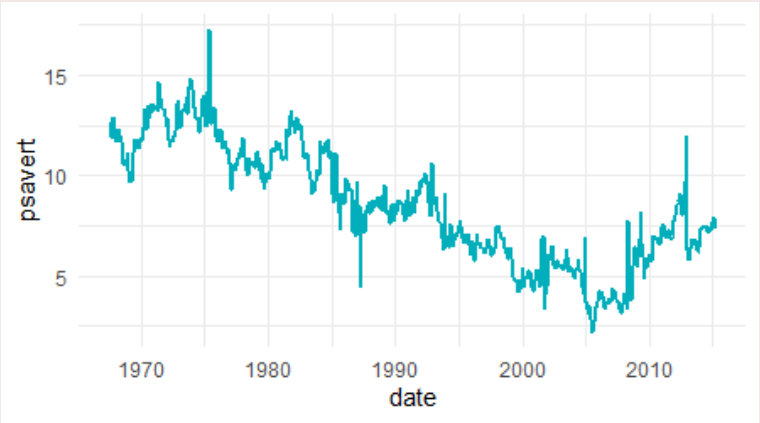
- Population Changes
- Technology Changes
- Inflation of Deflation (Price Changes) etc

Example of an Upward Trend



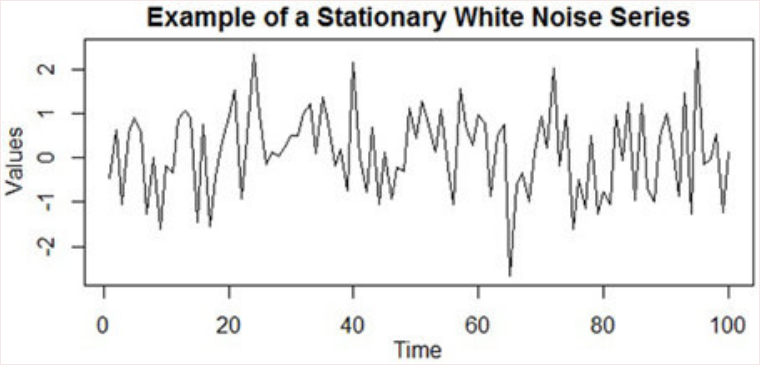
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Example of a Downward Trend



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Example of a Stationary Trend



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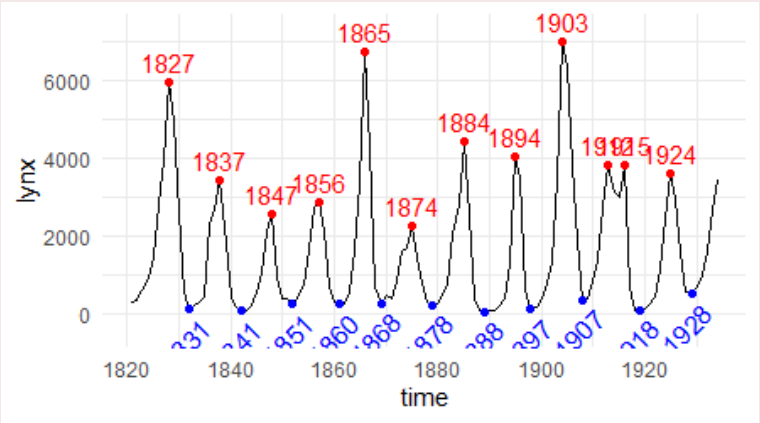
Cyclical Variation

Observable up and down fluctuations over an extended period of time. It could be as a result of a boom in business or bust in an activity.

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Example of a Cyclical Variation



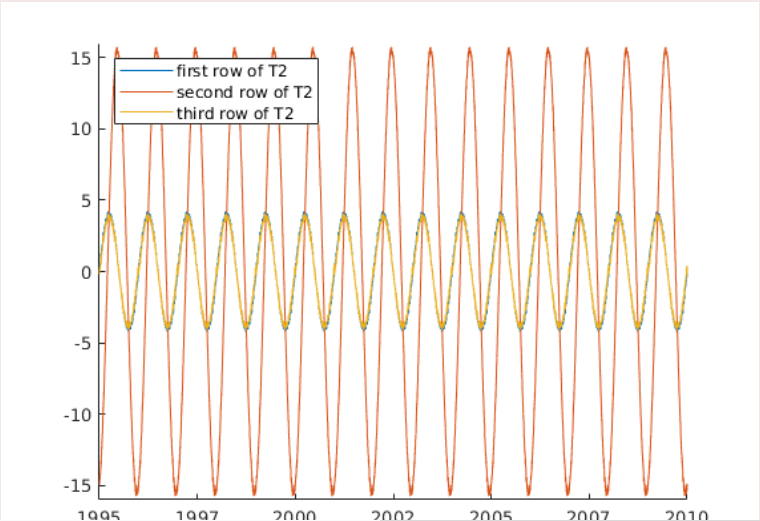
Seasonal Variation

This a variation that happens at a particular period of the year as a result of a particular event. It is caused by such factors as weather, customs etc.

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Example of a Seasonal Variation



Irregular Variation

These are variations which are erratic in movement over time. They are either unpredictable or caused by isolated events such as floods, earthquakes, government policy, etc.

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THANK YOU

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